



Dr. N.G.P Institute of Technology, Coimbatore- 48
(An Autonomous Institution)

22UCS402 – DATABASE MANAGEMENT SYSTEMS

UNIT- I DATABASE FUNDAMENTALS

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Introduction – Database System Applications – Views of data – Data models – Database languages – Database system architecture – Introduction to relational databases – Relational Model – Structure of Relational Databases – Database Schema – Keys – Relational Query Languages – Relational algebra

UNIT I DATABASE FUNDAMENTALS	
1.1	Introduction
1.2	Views of data – Data models
1.3	Data Models, Database languages
1.4	Database system architecture
1.5	Introduction to relational databases
1.6	Relational Model - Structure of Relational Databases
1.7	Database Schema – Keys
1.8	Relational Query Languages
1.9	Relational algebra

Agenda

- What is Data?
- Types of Data
- Structured Data and Unstructured Data
- How do we store Data into Computers ?
- File System Vs Database System
- File System Issues
- What is Management?
- What is DBMS?

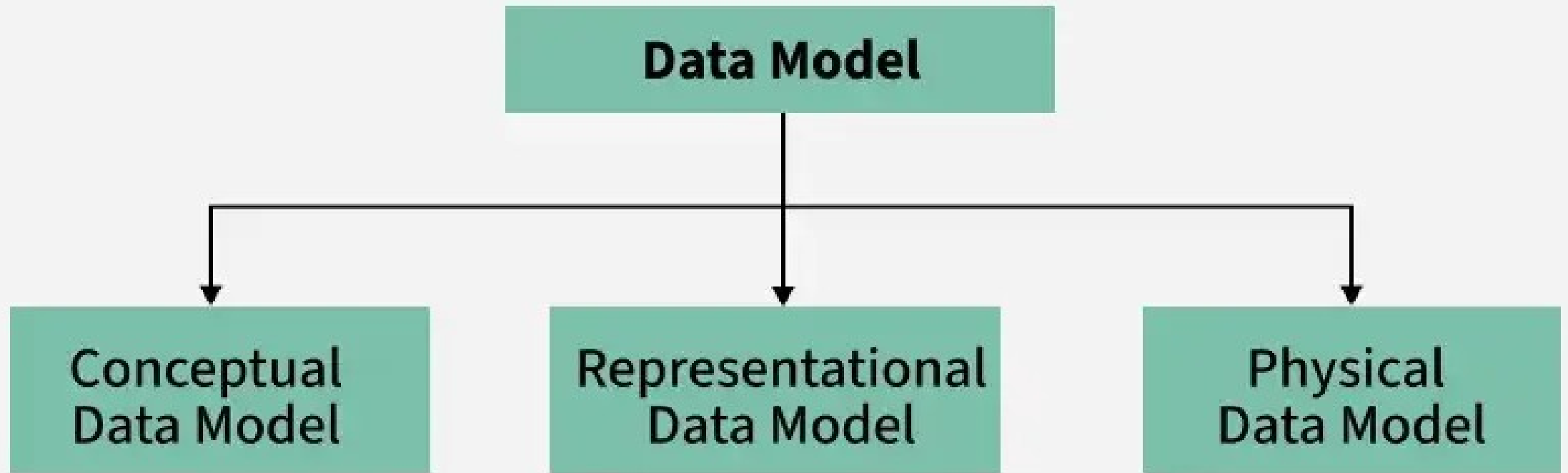
Data Models in DBMS

- A Data Model in Database Management System (DBMS) is the **concept of tools that are developed to summarize the description of the database.**
- Data Models provide us with a **transparent picture of data** which helps us in creating an actual database.
- **It shows us from the design of the data to its proper implementation of data.**

Types of Relational Models

- Conceptual Data Model
- Representational Data Model
- Physical Data Model

Data Models in DBMS



1. Conceptual Data Model

- The conceptual data model **describes the database at a very high level** and is useful to understand the **needs or requirements of the database**.
- It is this model, that is used in the requirement-gathering process i.e. **before the Database Designers start making a particular database**.
- One such popular model is the [entity/relationship model \(ER model\)](#).
- The E/R model specializes in **entities, relationships, and even attributes** that are used by database designers.

Entity-Relationship Model(ER Model):

- **Entity-Relationship Model(ER Model):** It is a high-level data model which is used to define the data and the relationships between them. It is basically a conceptual design of any database which is easy to design the view of data.

Components of ER Model:

- **Entity:** An entity is referred to as a **real-world object**. It can be a **name, place, object, class**, etc. These are represented by a **rectangle** in an ER Diagram.
- **Attributes:** An attribute can be defined as the **description of the entity**. These are represented by **Ellipse** in an ER Diagram. It can be **Age, Roll Number, or Marks** for a Student.
- **Relationship:** Relationships are used to **define relations** among different entities. **Diamonds and Rhombus** are used to show Relationships.

Characteristics of a conceptual data model

- Offers Organization-wide coverage of the business concepts.
- This type of Data Models are designed and developed for a **business audience**.
- The conceptual model is **developed independently of hardware specifications like data storage capacity, location or software specifications like DBMS vendor and technology**.
- The focus is to represent data as a user will see it in the “real world.”

2. Representational Data Model

- This type of data model is used to represent **only the logical part of the database** and does not represent the physical structure of the database.
- The representational data model allows us to focus primarily, on the **design part of the database**.
- A popular representational model is a [Relational model](#).
- The relational Model consists of [Relational Algebra](#) and [Relational Calculus](#).
- **In the Relational Model, we basically use tables to represent our data and the relationships between them. It is a theoretical concept whose practical implementation is done in Physical Data Model.**
- The advantage of using a Representational data model is to provide a foundation to form the base for the Physical model.

Characteristics of Representational Data Model

- Represents the **logical structure** of the database.
- Relational models like Relational Algebra and Relational Calculus are commonly used.
- **Uses tables** to represent data and relationships.
- Provides a foundation for building the physical data model.

3. Physical Data Model

- The physical Data Model is used to **practically implement Relational Data Model**.
- Ultimately, all **data** in a database is **stored physically** on a secondary storage device such as **discs and tapes**.
- This is stored in the form of **files, records**, and certain other data structures. It has all the information on the format in which the files are present and the structure of the databases, the presence of external data structures, and their relation to each other.
- Here, we basically **save tables in memory** so they can be accessed efficiently.

3. Physical Data Model

- In order to come up with a good physical model, **we have to work on the relational model in a better way.**
- [Structured Query Language \(SQL\)](#) is used to **practically implement Relational Algebra.**
- This Data Model describes **HOW** the system will be implemented using a specific DBMS system.
- This model is typically **created by DBA and developers.** The purpose is actual implementation of the database.

Characteristics of a physical data model:

- The physical data model describes data need for a single project or application though it maybe integrated with other physical data models based on project scope.
- Data Model contains **relationships between tables that which addresses cardinality and nullability of the relationships.**
- Developed for a specific version of a DBMS, location, data storage or technology to be used in the project.
- **Columns should have exact datatypes, lengths assigned and default values.**
- **Primary and Foreign keys, views, indexes, access profiles, and authorizations, etc. are defined**

Some Other Data Models- 1. Hierarchical Model

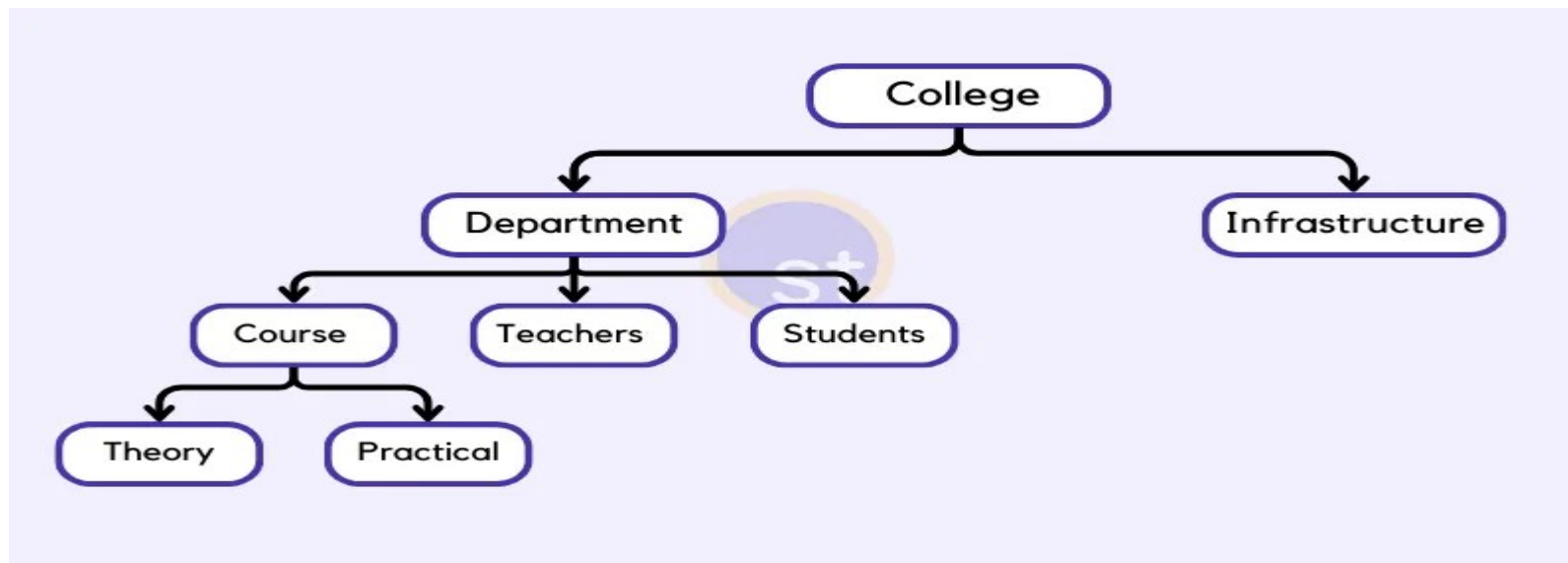
- The hierarchical database model organizes data into a **tree-like structure**, with a **single root**, to which all the other data is linked.
- The hierarchy starts from the **Root** data, and expands like a tree, adding **child** nodes to the **parent** nodes.
- In this model, a child node will only have a **single parent node**.
- This model efficiently describes many real-world relationships like the *index of a book*, etc.
- IBM's Information Management System (IMS) is based on this model.
- Data is organized into a tree-like structure with a **one-to-many relationship** between two different types of data, for example, one **department** can have many **courses**, many **teachers**, and of course many **students**(like shown in the diagram below).

Some Other Data Models- 1. Hierarchical Model

Advantages/Disadvantages of the Hierarchical Model

Here are a few points to mark the advantages and disadvantages of the Hierarchical database model:

- Because it has **one-to-many relationships** between different types of data so **it is easier and fast to fetch the data**.
- But the Hierarchical model is **less flexible**.
- And it doesn't support many-to-many relationships.



2. Network Model

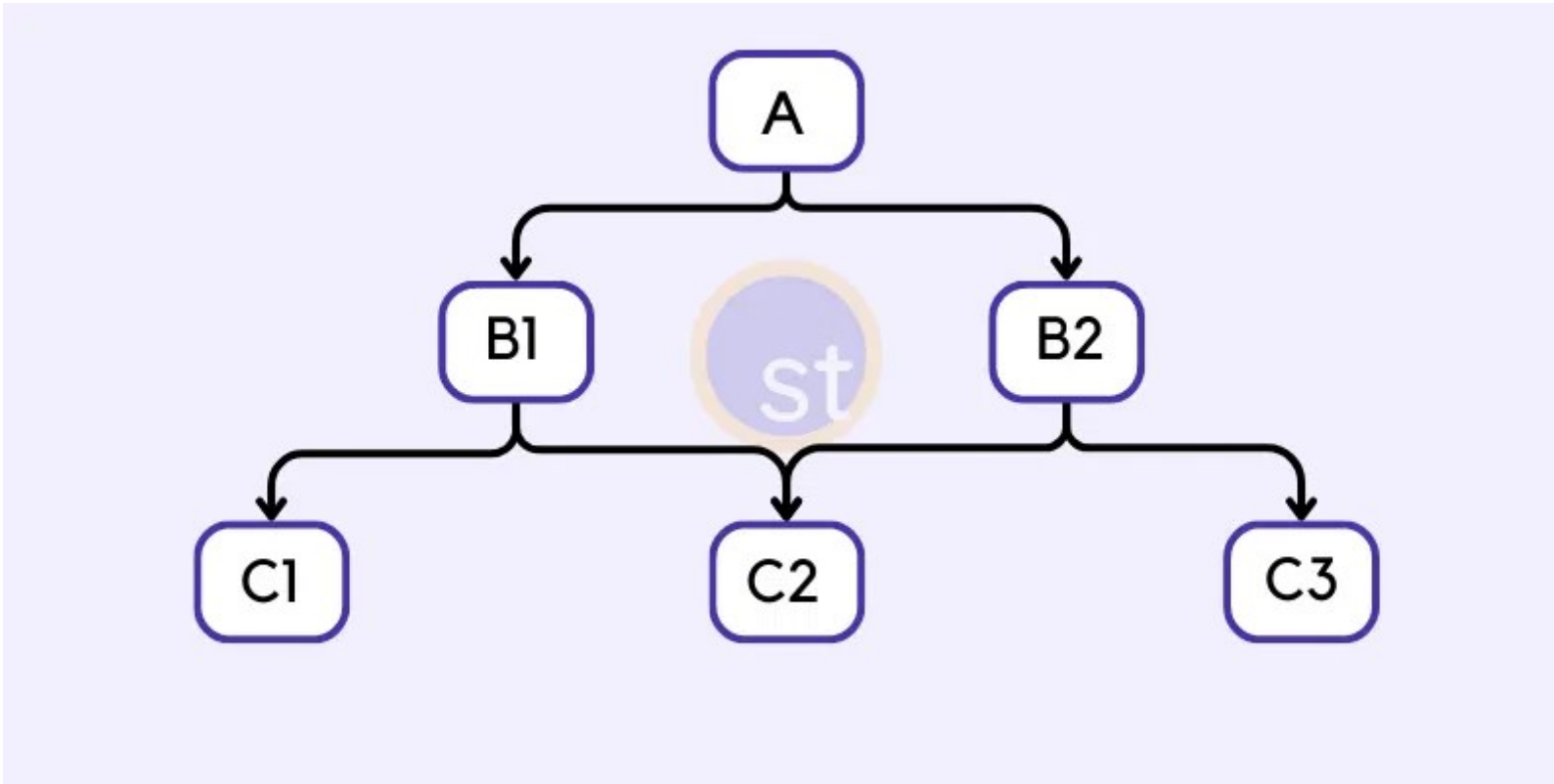
- The Network Model is an **extension of the Hierarchical model**.
- In this model, data is organized more like a **graph**, and allowed to have *more than one parent node*.
- In the network database model, data is **more related** as more relationships are established in this database model.
- Also, as the data is more related, hence **accessing the data is also easier and fast**.
- This database model uses many-to-many data relationships.
- **Integrated Data Store (IDS)** is based on this database model.

2. Network Model

- This was the **most widely used database model** before **Relational Model** was introduced.
- The **implementation** of the Network model is **complex**, and it's very difficult to maintain it.
- The Network model is **difficult to modify also**.
- You may want to explore this if you are developing some **social networking applications**, although the Graph Database model is new and is far better than the Network Database model.

Advantages of the Network Model

- It supports complex relationships
- It allows more flexibility



3. Entity-relationship Model

- In this database model, relationships are created by dividing **objects of interest into entities and their characteristics into attributes**.
- Different entities are related using **relationships**.
- ER Models are defined to represent the **relationships in pictorial form** to make it easier for different stakeholders to understand.
- This model is good to design a database, which can then be turned into tables in a relational model (explained below).

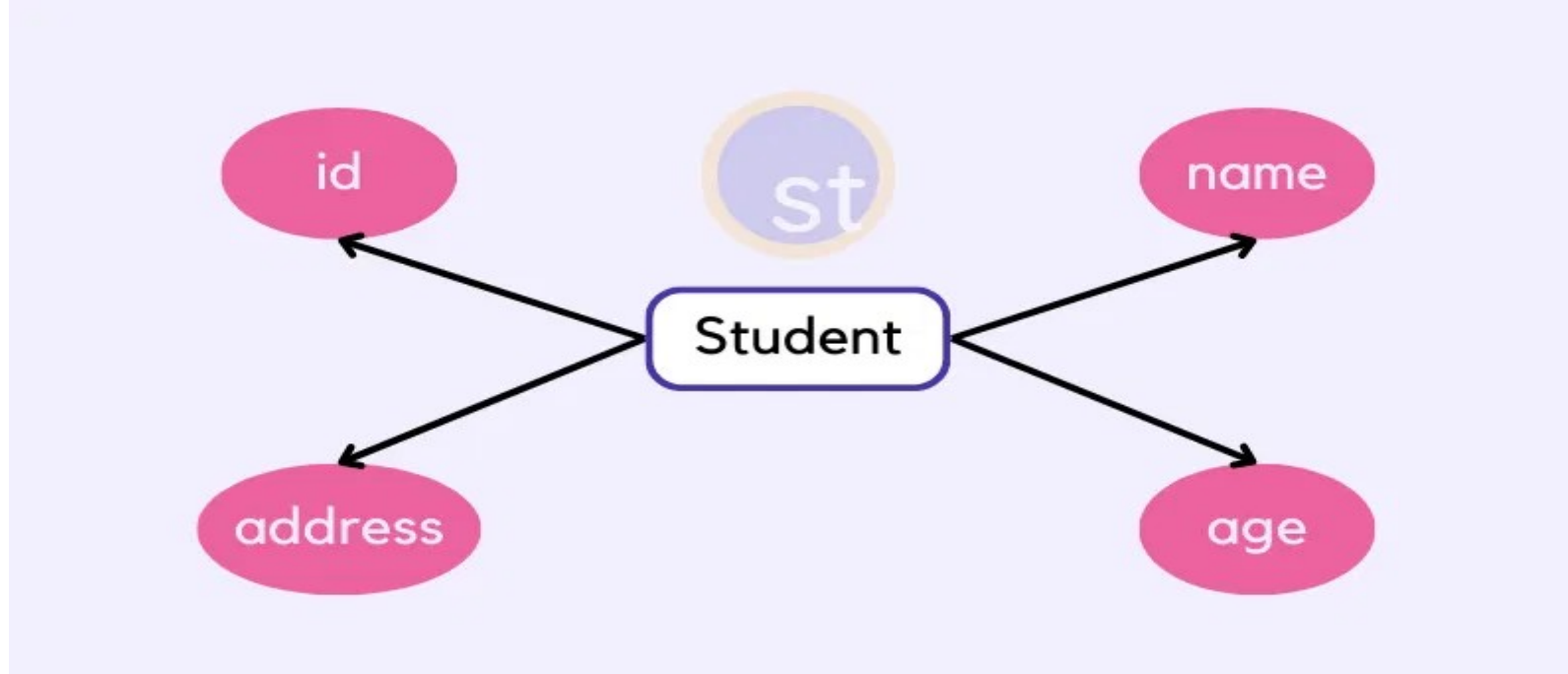
3. Entity-relationship Model

- Let's take an **example**, If we have to design a School Database, then the **Student** will be an **entity** with **attributes** *name, age, address*, etc. As an **Address** is generally complex, it can be another **entity** with **attributes** *street, pincode, city*, etc, and there will be a relationship between them.
- Relationships can also be of different types. You can learn about [ER Diagrams](#) in detail if you want to learn about entities and relationships.

3. Entity-relationship Model

Advantages of the ER Model

- It is easy to understand and design.
- Using the ER model **we can represent data structures easily.**
- As the ER model cannot be directly implemented into a database model, it is just a **step toward designing the relational database model.**



4. Relational Model

- In this model, data is organized in two-dimensional **tables** and the relationship is maintained **by storing a common field**.
- This model was introduced by **E.F Codd** in 1970, and since then it has been the most widely used database model.
- The basic structure of data in the relational model is **tables**. All the information related to a particular type is stored in **rows** of that table.
- Hence, tables are also known as **relations** in the relational model.
- You can design tables, [normalize them](#) to reduce data redundancy, and use [Structured Query language or SQL](#) to access data from the tables.
- Some of the most popular databases are based on this database model. For example, **Oracle**, **MySQL**, etc.

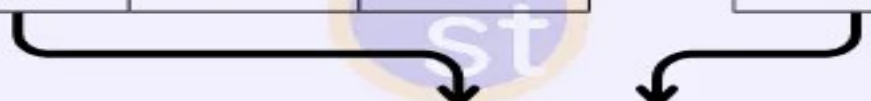
4. Relational Model

Advantages of the Relational Model

- It's **simple and easy** to implement.
- Popular database software is available for this database model.
- It supports SQL using which you can **easily query the data**.

student_id	name	age
1	Akon	17
2	Bkon	18
3	Ckon	17
4	Dkon	19

subject_id	name	teacher
1	Java	Mr. J
2	Python	Miss Py
3	JavaScript	Mrs. JS
4	C#	Mr. C



student_id	subject_id	marks
1	1	98
1	2	77
2	1	76
3	2	88

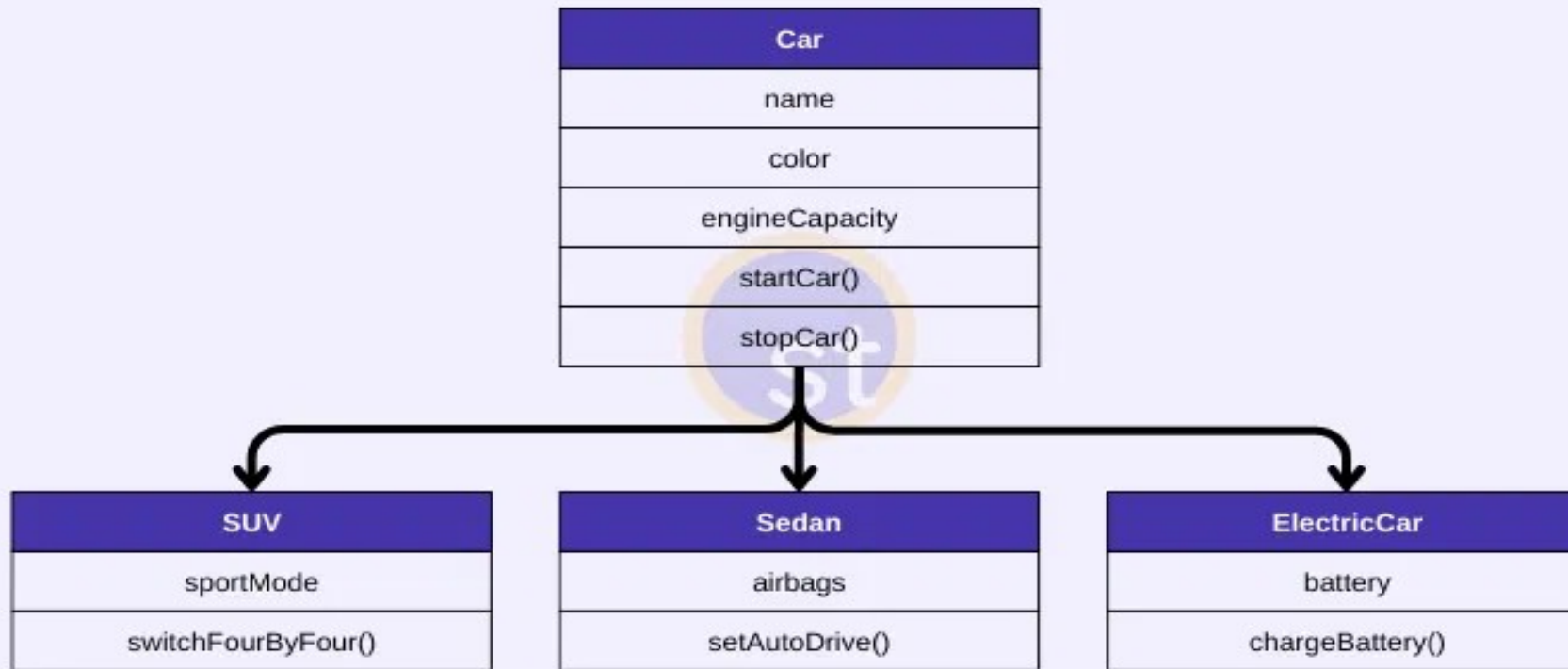
5. Object-oriented Model

- In this model, data is stored in the **form of objects**.
- The behavior of the object-oriented database model is just like object-oriented programming.
- A very popular example of an Object Database management system or **ODBMS** is **MongoDB** which is also a NoSQL database.
- This database model is not mature enough as compared to the relational database model.

5. Object-oriented Model

Advantages of the Object-oriented Model

- It can easily support complex data structures, with relationships.
- It also supports features like Inheritance, Encapsulation, etc.



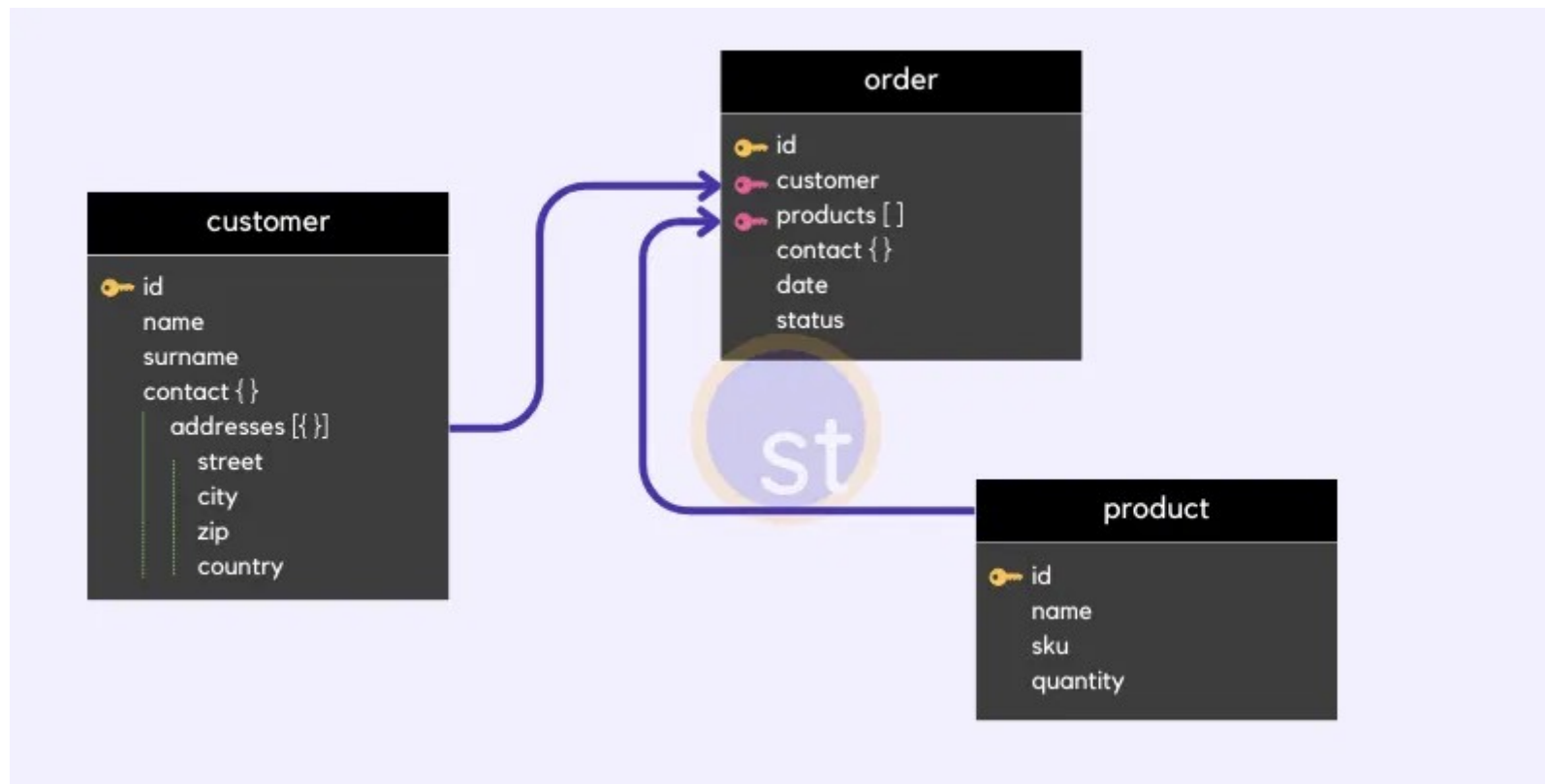
6. NoSQL Model

- The NoSQL database model supports an **unstructured** style of storing data.
- Data is stored as **documents**.
- The documents look more like **JSON strings** or **Key-value** based object representations.
- It provides a **flexible schema**.
- It does provide features like **indexing**, relationships between data, etc.
- The support for **data querying is limited** in the NoSQL database model.
- This database model is well-suited for **Big data applications, real-time analytics, CMS** (Content Management systems), etc.

6. NoSQL Model

Advantages of the NoSQL Model

- This database model is scalable.
- This database model functions with high performance.
- The NoSQL database model can **handle large volumes of data**.



7. Graph Model

- The Graph database model is based on more **real-world like relationships**.
- Data is represented using **Nodes** or entities.
- The nodes are related using **edges**.
- The popular database **Neo4j** is based on the Graph database model.
- If your application has simple data requirements, then you should not use the graph database model.
- In modern applications like **social networks, recommendation systems**, etc. the graph database model is well-suited.

7. Graph Model

Advantages of the Graph Model

- It handles complex relationships very well.
- In the modern world where there is so much data and the data has to be related in different ways, the graph database model is very useful.

